King's College London

This paper is part of an examination of the College counting towards the award of a degree. Examinations are governed by the College Regulations under the authority of the Academic Board.

Examination Period	August 2024 (Period 3)
Module Code	6CCS3HCI
Module Title	Human-Computer Interaction

Format of Examination	Written questions
Start time	11am BST (GMT +1)
Time Allowed	Two hours
Instructions	You are permitted to access any materials you wish, but this is not mandated and is not expected. You may use a calculator if you find this helpful.
Rubric	ANSWER ALL QUESTIONS

The rubric for this paper must be followed and extra answers should not be submitted. For answers that are handwritten, write with blue/black ink on light coloured paper. Include the Module code, question number and student number on every page to be submitted. For answers that are typed, use the template provided.

Submission Deadline	
Submission Process	Work must be submitted to the **level 6** Informatics Assessments KEATS page.

Your work must be submitted as a PDF file. If you have prepared some answers on computer, and some on paper (which have then been digitised), you may upload at most two PDF files – one for computer-prepared answers, one for digitised answers. Do not duplicate answers across the two PDFs – if you do this, the computer-prepared answer will be taken. You should check that your work displays correctly after it has been uploaded.

ACADEMIC HONESTY AND INTEGRITY: Students at King's are part of an academic community that values trust, fairness and respect and actively encourages students to act with honesty and integrity. It is a College policy that students take responsibility for their work and comply with the university's standards and requirements.
By submitting this assignment, I confirm that this work is entirely my own, or is the work of an assigned or permitted group, of which I am a member, with exception to any content where the works of others have been acknowledged with appropriate referencing. I also confirm that I have read and understood the College's Academic Honesty & Integrity Policy:
`https://www.kcl.ac.uk/governancezone/assessment/academic-honesty-integrity`
Misconduct regulations remain in place during this period and students can familiarise themselves with the procedures on the College website at
`https://www.kcl.ac.uk/campuslife/acservices/academic-regulations/assets-20-21/g27.pdf`

Important note: All the questions within this exam will refer to your design context, specified below. Please make sure to clearly connect your responses to this design problem – general descriptions of design methods or approaches will not be marked unless they also directly explain how the given method is to be applied to your problem.

For example, when answering a question on prototyping, describing only the general process of how you would prototype (select an assumption to test, identify low/high-fidelity method ...) would lead to 0 marks unless also accompanied by specific examples of assumptions that would be meaningful in your context, and the specific choices you could make for your prototype.

Your design context:

Your company has been hired by TrustyTrader to design an app that helps people find workers to help with tasks around the home (e.g. cleaning, assembling furniture, plumbing). The specific focus of your user experience team is to make sure that the app works the best it can for people who have never used an app like this before.

1. **Empathise – envision the needs and challenges of your users.**

   a. Imagine you are to interview 15 future users from your population. Assume you have already consent and have introduced the interviewee to the topic.

   i. Explain who you would recruit for this experiment and why.

   ii. Draft an interview guide you could use, including 8-10 key questions you would ask.

   iii. Outline the key areas these questions will be covering – i.e., what aspects of the users' experience are you trying to learn more about and why.

   [17 marks]

   b. Sketch an empathy map diagram and fill it out with 3 observations for each quadrant that you could imagine emerging from the interviews. Do not worry about what exactly your imagined observations say; we are trying to see how you are able to think about the aims of the empathy map process, rather than how well you imagine what 'real' users would say.

   [8 marks]

2. **Define – What is your problem statement, and why is it the most interesting to design for?**

   a. Drawing on your key areas from your interview and the envisioned 'empathy map', make a simple persona depicting a member of your target population.

   [10 marks]

   b. Drawing on your persona, create a meaningful problem statement, using one of the templates from the lectures. Explain why you selected this statement, and why is it likely important for your envisioned users?

   [10 marks]

3. Ideate & Prototype – what assumptions do you envision to be key in this design context?

a. Generate 5 HMW questions that you see plausible in your design context.

[5 marks]

b. Select one of the HMW questions above to 'prototype'.

i. Identify and describe a key design assumption that you would need to test first.

ii. Explain why this is the most important assumption in your design context.

[10 marks]

c. For that key design assumption from point b:

i. What prototyping methods would be most useful to test it? For example with which fidelity would you prototype? With how much researcher involvement? With what technologies?

ii. Explain your decisions.

[10 marks]

4. Evaluate – how would you know that your system works?

a. Going back to your problem statement (and assuming your prototyping was successful), outline what would be useful evaluation metrics for your system. Consider:

i. What does 'works better' mean in this particular context?

ii. Identify at least 3 qualitative or quantitative indicators that you could use as a measure of success of your app.

iii. Explain why you selected these indicators, and how they might be measured.

[10 marks]

b. Outline two evaluation approaches you could use to answer the question in the context of this design problem. One should be more 'in the wild', and the other should be more controlled.

i. Describe the rationale for each choice.

ii. What questions would **not** be answered by each approach?

[10 marks]

c. Pick one of these approaches that you think would be most useful, and explain how you would run a full study in this context.

i. How many users would you aim to recruit and why?

ii. List and briefly explain key steps in the study protocol.

[10 marks]